

Photoelectric Effect Lab

Han Fang and Rooney Cheung

March 16, 2026

Abstract

This lab investigated Einstein's theory of the photoelectric effect. It was determined that the stopping voltage was proportional and strongly correlated with the frequency of light while also independent of the intensity of light, and that the photocurrent was proportional and strongly correlated with the intensity of light, confirming theory. However, it was also found that the stopping voltage was proportional to the intensity of light, which contradicts theory. Furthermore, the experimental Planck's constant was $(4.9 \pm 0.4) \times 10^{-34} [Js]$, the work function of the metal was $(1.4 \pm 0.2) \times 10^{-19} [J]$, and the threshold frequency was $(2.9 \pm 0.5) \times 10^{14} [Hz]$. These findings may have been affected by sources of error, such as bandwidths of each LED, low intensity measurements and noise from the background.

Introduction

Photoelectric Effect

In 1899, J.J. Thomson discovered that when a metal target is irradiated by UV light, the metal target emits photoelectrons in response. This phenomenon is governed by:

$$E_k = E_e - E_o \quad (1)$$

where $E_k [J]$ is the electron kinetic energy, $E_e [J]$ is total energy absorbed by electrons, $E_o [J]$ is the work function, dependent on the metal property.

After Thomson's initial discovery, Phillip Lenard conducted an experiment using a vacuum chamber with two electrodes, separated by a potential difference, $\Delta V [V]$. A light source with frequency greater than the threshold frequency, $f_0 [s^{-1}]$ is shone on the cathode surface to generate a photocurrent, $I_{photo} [A]$. As a result, the stopping voltage, $V_{stop} [V]$, defined as the voltage where an electron with kinetic energy E_k would be rejected from the anode is given by

$$V_{stop} = \frac{E_k}{e} \quad (2)$$

Where e is the elementary charge constant.

In addition, Albert Einstein's discovery of the quantization of electromagnetic radiation in 1905, states that:

$$E = hf \quad (3)$$

where $h = 6.63 \times 10^{-34} Js$ is Planck's Constant, $f = c/\lambda [s^{-1}]$ is the frequency of radiation.

Combining Equations 1, 2, 3, the stopping voltage can be rewritten as:

$$V_{stop} = \frac{h}{e}f - \frac{E_0}{e} \quad (4)$$

Or, in terms of a threshold frequency f_0 ,

$$V_{stop} = \frac{h}{e}(f - f_0) \quad (5)$$

Note that this indicates that V_{stop} is independent of the intensity of the light shone, while photocurrent is proportional to the intensity of the light shown, highlighted by Maxwell's equations. In this experiment, the photocurrent is not directly determined, but instead obtained from a voltage measurement, $V_{photo} [V]$, across a known resistance $R = 100k\Omega$, this is given by:

$$I_{photo} = \frac{V_{photo}}{R} \quad (6)$$

Uncertainties and Uncertainty Propagation

Uncertainties are a measure of how precise the measurement or calculation of a variable is. In this experiment, due to the high accuracy of the multimeter and the large spread of voltage value, statistical uncertainty (standard deviation, ΔV) for V_{photo} and V_{stop} dominates, given by:

$$\Delta V = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\bar{V} - V_i)^2} \quad (7)$$

where N is the number of data points, $\bar{x}$ is the average value of the data points, and x_i is the i th data point.

In general, the uncertainty of constant variable product can be expressed as follows [1]:

$$\Delta y = c\Delta x \quad (8)$$

Where x and y are variables with uncertainties denoted with Δ , c is a constant, and $y = cx$.

Furthermore, the uncertainty of one variable divided by another variable can be expressed as

$$\Delta z = \frac{x}{y} \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2} \quad (9)$$

Where x , y and z are variables with uncertainties denoted with Δ , and $z = x/y$ [1].

The photocurrent uncertainty, ΔI_{photo} , can thus be propagated from the resistor tolerance, $\Delta R = 100k\Omega \times 5\% = 5k\Omega$ and ΔV_{photo} in accordance with Eqn. 6 and 9:

$$\Delta I_{photo} = \frac{V_{photo}}{R} \sqrt{\left(\frac{\Delta V_{photo}}{V_{photo}}\right)^2 + \left(\frac{\Delta R}{R}\right)^2} \quad (10)$$

In addition, the uncertainty for frequencies, Δf [s^{-1}] are inferred from the bandwidth of each LED color, this is given by:

$$\Delta f = \frac{c}{\lambda^2} \Delta \lambda \quad (11)$$

where λ is the wavelength, $\Delta \lambda$ is half of the bandwidth labeled on the manufacturer's technical specification respectively, $c = 299,792,458m/s$ is the speed of light.

Finally, the intensity ratio uncertainty, Δi , controlled by a knob (See Procedures), is estimated to be 1/2 the ratio of an interval length to the length of the full dial from 0 to the limit, given by:

$$\Delta i = \frac{1}{2} \times \frac{1}{6} \approx 0.08 \quad (12)$$

Experimental Method

Materials

- 2 Digital Multimeters
- A Control circuit, with a 9 Volt Source, $10k\Omega$, $100k\Omega$, $2.5k\Omega$ resistor each.
- Phototube
- 8 LEDs of different wavelengths
- LED Power Supply Circuit

Procedures

Setup

1. Connect the circuit as shown below in the circuit diagram in Figure 1, ensure the multimeters are set to voltage mode, each measuring V_{stop} and V_{photo} respectively.
2. Once the circuit is set up, select the Red LED, connect it to the Power Supply circuit and insert it into the phototube, the complete setup is shown in Figure 2:
3. Change the mode of the LED Power supply to CW (constant voltage).

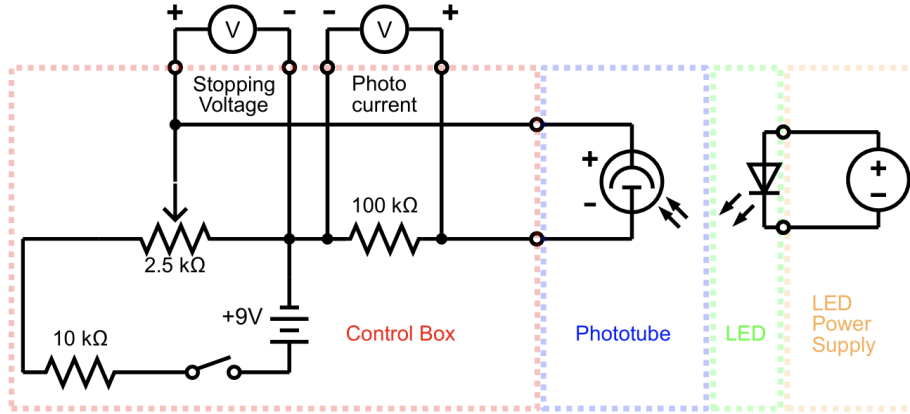


Figure 1: Circuit diagram for the Lab [2]

Part I

- Adjust the potentiometer until you obtain a V_{stop} that produces a zero V_{photo} (and by extent a zero I_{photo}), record a range of 5 instantaneous values of the respective V_{stop} .
- Similarly, adjust the potentiometer until you obtain a V_{photo} that produces a zero V_{stop} , record a range of 5 instantaneous values of the respective V_{photo} .
- Change the LED, repeat the previous instructions of Part I until all 8 LEDs have data recorded.

Part II

- Pick any LED (*In our case, we picked Green*).
- Using that LED, vary the intensity of the LED by rotating the knob on the Power Supply of the LED to the designated ticks (0.17, 0.33, 0.50, 0.67, 0.83 and 1) that represent the ratio between the selected light intensity through the knob and the maximum possible intensity, marked by a tick on the rotating knob labelled "LIMIT". For each intensity ratio, record the respective range of 5 V_{photo} and V_{stop} values.
- Repeat until the preset intensity values all have associated data.

Raw Experimental Data

Part I

This part includes V_{stop} and I_{photo} when the other respective variable is 0, also includes the frequencies and bandwidths of each LED colour.

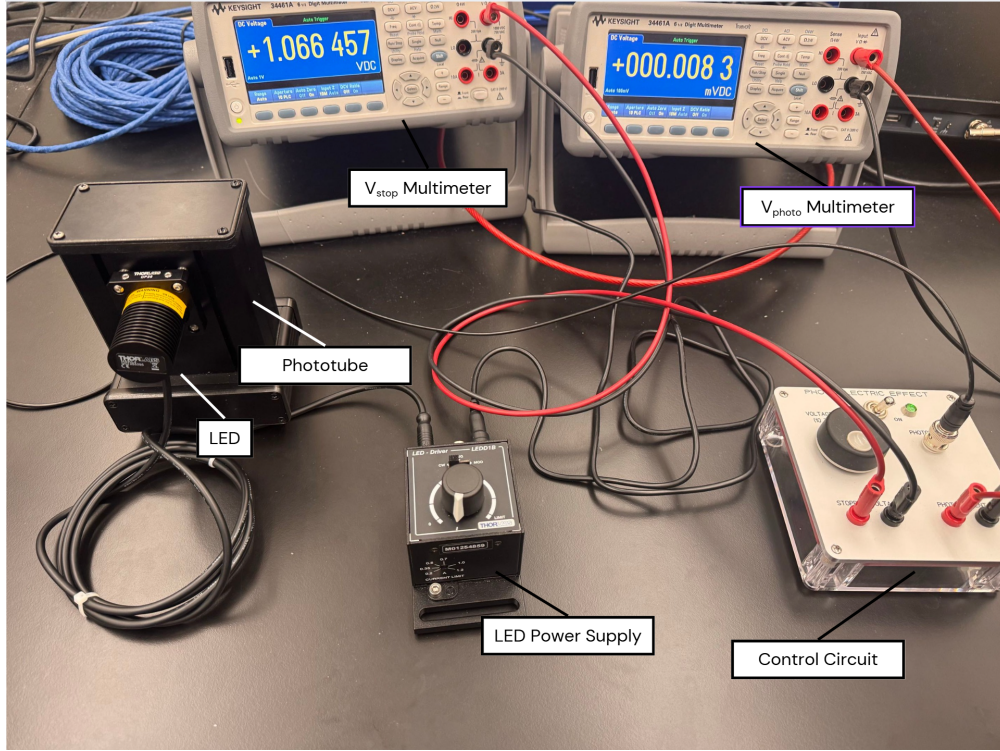


Figure 2: Setup of the lab, note that the LED is placed inside the phototube

LED	$V_{stop,1}$	$V_{stop,2}$	$V_{stop,3}$	$V_{stop,4}$	$V_{stop,5}$	$V_{stop,avg}$ [V]	$\pm\Delta V_{stop}$ [V]
Violet	1.4376	1.4363	1.4358	1.4375	1.4366	1.4368	0.0008
Blue	1.1295	1.1294	1.1325	1.1301	1.1293	1.1301	0.0014
Cyan	0.8716	0.8702	0.8723	0.8717	0.8723	0.8716	0.0008
Green	0.8138	0.8137	0.8136	0.8137	0.8137	0.8137	0.0001
Amber	0.6145	0.6165	0.6150	0.6120	0.6157	0.6147	0.0017
Orange	0.4887	0.4879	0.4879	0.4885	0.4896	0.4885	0.0007
Red	0.3870	0.3861	0.3872	0.3862	0.3875	0.3868	0.0006
Infrared	0.1796	0.2040	0.2018	0.1952	0.2278	0.2017	0.0174

Table 1: Stopping Potential across five trials, including average and uncertainty for all 8 LEDs. The uncertainty of the stopping voltage ΔV_{stop} was determined by Eqn. 7.

LED	λ [nm]	Bandwidth, $2\Delta\lambda$ [nm]	f [s^{-1}]	$\pm\Delta f$ [s^{-1}]
Violet	405	15	7.4×10^{14}	1×10^{13}
Blue	460	15	6.5×10^{14}	1×10^{13}
Cyan	515	30	5.8×10^{14}	2×10^{13}
Green	540	40	5.6×10^{14}	2×10^{13}
Amber	595	15	5.04×10^{14}	6×10^{12}
Orange	625	15	4.80×10^{14}	6×10^{12}
Red	660	20	4.54×10^{14}	7×10^{12}
Infrared	945	20	3.17×10^{14}	3×10^{12}

Table 2: Wavelengths, bandwidths, and frequencies of all 8 LEDs obtained from the Manufacturer’s Technical Specification. The uncertainty of the light frequency Δf was determined by Eqn. 11.

LED	$V_{photo,1}$	$V_{photo,2}$	$V_{photo,3}$	$V_{photo,4}$	$V_{photo,5}$	V_{avg} [V]	$\pm\Delta V_{photo}$ [V]	I_{photo} [A]	$\pm\Delta I_{photo}$ [A]
Violet	4.4×10^{-1}	4.4×10^{-1}	4.4×10^{-1}	4.4×10^{-1}	4.4×10^{-1}	4.43×10^{-1}	3×10^{-5}	4.4×10^{-6}	2×10^{-7}
Blue	3.5×10^{-1}	3.5×10^{-1}	3.5×10^{-1}	3.5×10^{-1}	3.5×10^{-1}	3.49×10^{-1}	5×10^{-6}	3.5×10^{-6}	2×10^{-7}
Cyan	2.3×10^{-1}	2.3×10^{-1}	2.3×10^{-1}	2.3×10^{-1}	2.3×10^{-1}	2.31×10^{-1}	5×10^{-6}	2.3×10^{-6}	1×10^{-7}
Green	1.9×10^{-1}	1.9×10^{-1}	1.9×10^{-1}	1.9×10^{-1}	1.9×10^{-1}	1.87×10^{-1}	1×10^{-5}	1.9×10^{-6}	9×10^{-8}
Amber	8.6×10^{-2}	8.6×10^{-2}	8.6×10^{-2}	8.6×10^{-2}	8.6×10^{-2}	8.59×10^{-2}	1×10^{-5}	8.6×10^{-7}	4×10^{-8}
Orange	6.2×10^{-2}	6.2×10^{-2}	6.2×10^{-2}	6.2×10^{-2}	6.2×10^{-2}	6.18×10^{-2}	9×10^{-6}	6.2×10^{-7}	3×10^{-8}
Red	2.7×10^{-2}	2.7×10^{-2}	2.7×10^{-2}	2.7×10^{-2}	2.7×10^{-2}	2.68×10^{-2}	5×10^{-6}	2.7×10^{-7}	1×10^{-8}
Infrared	8.7×10^{-6}	1.0×10^{-5}	1.6×10^{-5}	8.1×10^{-6}	2.2×10^{-5}	1.40×10^{-5}	6×10^{-6}	1.4×10^{-10}	6×10^{-11}

Table 3: Photovoltage and photocurrent across 5 trials, including average and uncertainty for all 8 LEDs. The uncertainty of the photovoltage ΔV_{photo} was determined by Eqn. 7, and the photocurrent I_{photo} was determined by Eqn. 6 with its uncertainty ΔI_{photo} determined by 10.

Part II

This part includes V_{stop} and I_{photo} for the Green LED for 6 different intensity ratio (i) values.

i [± 0.08]	$V_{stop,1}$ [V]	$V_{stop,2}$ [V]	$V_{stop,3}$ [V]	$V_{stop,4}$ [V]	$V_{stop,5}$ [V]	$V_{stop,avg}$ [V]	$\pm \Delta V_{stop}$ [V]
0.17	0.567	0.567	0.567	0.567	0.567	0.5667	0.00003
0.33	0.567	0.567	0.567	0.567	0.567	0.5666	0.00002
0.50	0.567	0.567	0.567	0.567	0.567	0.5666	0.00002
0.67	0.567	0.567	0.567	0.567	0.567	0.5666	0.00007
0.83	0.567	0.567	0.567	0.567	0.567	0.5666	0.00002
1.00	0.567	0.567	0.567	0.567	0.567	0.5666	0.00001

Table 4: Stopping potential across five trials for 6 intensities, including average value and uncertainties. The uncertainty of the intensity ratio Δi was determined in Eqn. 12. Furthermore, the uncertainty of the stopping voltage ΔV_{stop} was determined by Eqn. 7.

i [± 0.08]	$V_{photo,1}$ [mV]	$V_{photo,2}$ [mV]	$V_{photo,3}$ [mV]	$V_{photo,4}$ [mV]	$V_{photo,5}$ [mV]	$V_{photo,avg}$ [mV]	$\pm \Delta V_{photo}$ [mV]	I_{photo} [A]	$\pm \Delta I_{photo}$ [A]
0.17	0.9641	0.8941	1.0189	0.9930	0.9649	0.97	0.05	9.7×10^{-9}	7×10^{-10}
0.33	4.4076	4.4427	4.4740	4.4461	4.4472	4.44	0.02	4.4×10^{-8}	2×10^{-9}
0.50	7.5025	7.5300	7.5137	7.5344	7.5459	7.53	0.02	7.5×10^{-8}	4×10^{-9}
0.67	16.9824	16.6838	17.0782	16.9745	16.8241	16.91	0.16	1.7×10^{-7}	9×10^{-9}
0.83	24.0740	24.3225	24.1433	24.2601	24.4279	24.25	0.14	2.4×10^{-7}	1×10^{-8}
1.00	27.1932	27.2081	27.2210	27.2429	27.2602	27.23	0.03	2.7×10^{-7}	1×10^{-8}

Table 5: Photovoltage and photocurrent across 5 trials, including average and uncertainty for all 6 intensities. The uncertainty of the intensity ratio Δi was determined in Eqn. 12. Furthermore, the uncertainty of the photovoltage ΔV_{photo} was determined by Eqn. 7, and the photocurrent I_{photo} was determined by Eqn. 6 with its uncertainty ΔI_{photo} determined by 10.

Results and Analysis

Part I

Stopping Voltage vs. Light Frequency

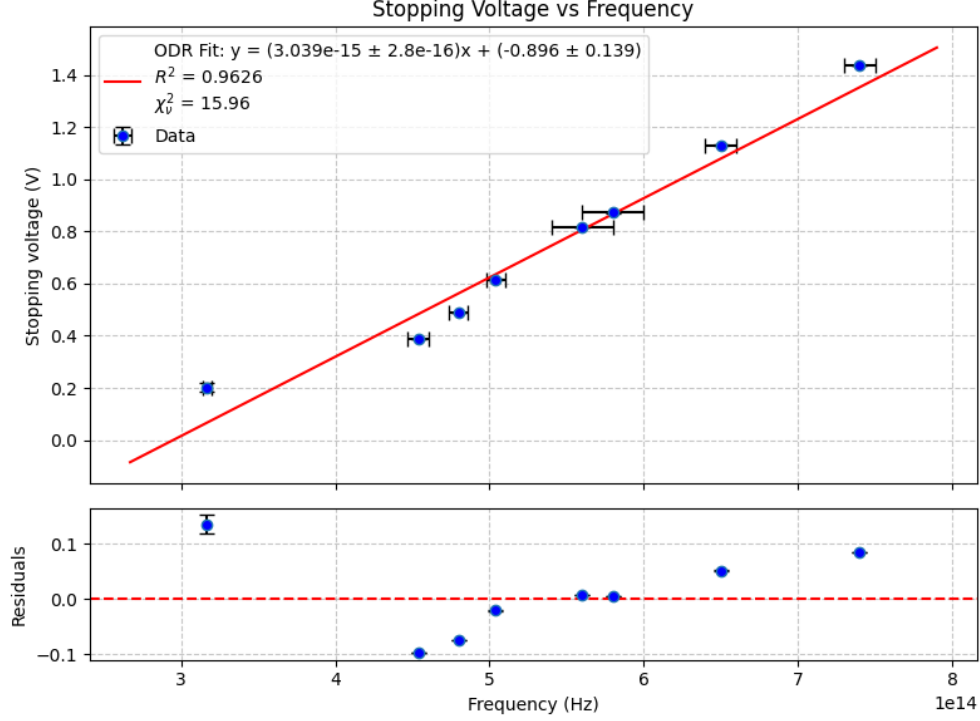


Figure 3: A graph of stopping voltage [V] versus frequency of light [Hz], based on Tables 1 and 2, with a corresponding plot of residuals. Error bars are present but may not be visible. The linear regression is $V_{stop} = [(3.0 \pm 0.3) \times 10^{-15} \text{Vs}]f + [(-0.9 \pm 0.1)\text{V}]$. $R^2 = 0.9626$, implying that the stopping voltage is strongly correlated to the frequency of light. However, $\chi^2_\nu = 15.96 > 1$, which implies that the uncertainties of each data point were underestimated. See the Sources of Error section for possible causes of such discrepancies.

From Fig. 3, the relationship between the stopping voltage V_{stop} [V] and the frequency of light f [Hz] was found to be

$$V_{stop} = [(3.0 \pm 0.3) \times 10^{-15} \text{Vs}]f + [(-0.9 \pm 0.1)\text{V}] \quad (13)$$

Comparing Eqn. 13 to Eqn. 4, the experimental Planck's constant h_{exp} [Js] is determined by multiplying the slope of Eqn. 13 by e , and the corresponding uncertainty value determined by Eqn. 8. Thus,

$$h_{exp} = (4.9 \pm 0.4) \times 10^{-34} [\text{Js}]$$

This experimental value is around 26% lower than the theoretical Planck's constant of $6.626 \times 10^{-34} [Js]$. The theoretical Planck's constant value is also more than 4 uncertainties away from the experimental value, implying that there were some systematic experimental errors. See the Sources of Error section for possible causes of such discrepancies.

Furthermore, comparing Eqn. 13 to Eqn. 4 also yields the work function $E_{0_{exp}}$ [J] of the metal in the phototube by multiplying the negative of the y-intercept of Eqn. 13 by e , with the corresponding uncertainty value determined by Eqn. 8. Thus,

$$E_{0_{exp}} = (1.4 \pm 0.2) \times 10^{-19} [J]$$

Combining the results for the experimental Planck's constant and work function, by Eqn. 5 the threshold frequency f_0 [Hz] is determined with $f_0 = E_{0_{exp}}/h_{exp}$, with the corresponding uncertainty value determined by Eqn. 9. Thus,

$$f_0 = (2.9 \pm 0.5) \times 10^{14} [Hz]$$

Photocurrent vs. Light Frequency

From Fig. 4, it seems that the photocurrent generated increases as the frequency of light increases. However, theory states that the photocurrent should be independent of the frequency of light, and instead increase with increased light intensity [3]. This specific error may be because the higher frequency LEDs in this experiment were manufactured to have greater power output, which would increase the light intensity, and thus the photocurrent, as LEDs with higher frequencies are used. Other possible errors are discussed in the Sources of Error section.

Part II

Stopping Voltage vs. Light Intensity

From Fig. 5, most of the data points are constant, with $V_{stop} = 0.5666V$, and one outlier that has just a $0.0001V$ difference from the mode. The slope of Fig. 5 is also less than 2 uncertainties away from 0, meaning that there is a non-negligible probability that the plot between stopping voltage and intensity is constant. Thus, it is confirmed that stopping voltage is independent of the light intensity.

Photocurrent vs. Light Intensity

From Fig. 6, the photocurrent generated is strongly correlated with light intensity such that the photocurrent increases linearly as the intensity increases. This result is consistent with theory, as more electrons are emitted by the phototube as the light intensity increases and more photons hit the phototube [3].

Sources of Error

One source of error throughout both parts of this experiment comes from the bandwidths of each LED present, especially the green LED. This large spread of wavelength leads to

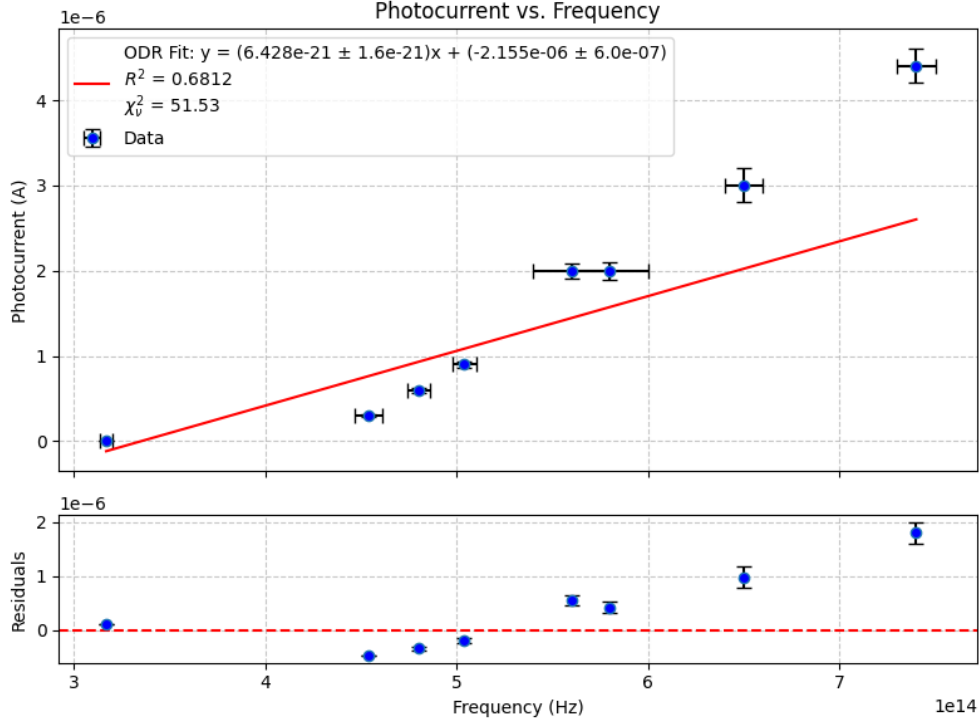


Figure 4: A graph of photocurrent [A] versus frequency of light [Hz], based on Tables 2 and 3, with a corresponding plot of residuals. Error bars are present but may not be visible. The linear regression is $I_{photo} = [(6 \pm 2) \times 10^{-21} \text{As}]f + [(-2.2 \pm 0.6) \times 10^{-6} \text{A}]$. $R^2 = 0.6812$, implying that the fit between photocurrent and frequency of light is mediocre. In addition, $\chi^2_\nu = 51.53 > 1$, which implies that the uncertainties of each data point were underestimated. See the Sources of Error section for possible causes of such discrepancies.

varying frequency values as opposed to the sole value corresponding to the peak wavelength. This introduces discrepancies in our fits as the frequency cannot be accurately determined.

Another source of error would be the accurate determination of V_{stop} and V_{photo} when the other is at 0. This is especially difficult in low intensities in Part II, as shown in Table 4, the stopping voltage at low intensity hugely deviates from the rest of the data.

Finally, the noise at the background introduces random noise into our data, this is due to the resistor's fluctuating voltage value at near 0-Volts condition as well as the multimeter's underlying noise. Another source of noise would be light seeping into the phototube at the small openings. Though these sources are addressed through taking averages across different trials, nonetheless the statistical deviation still persists.

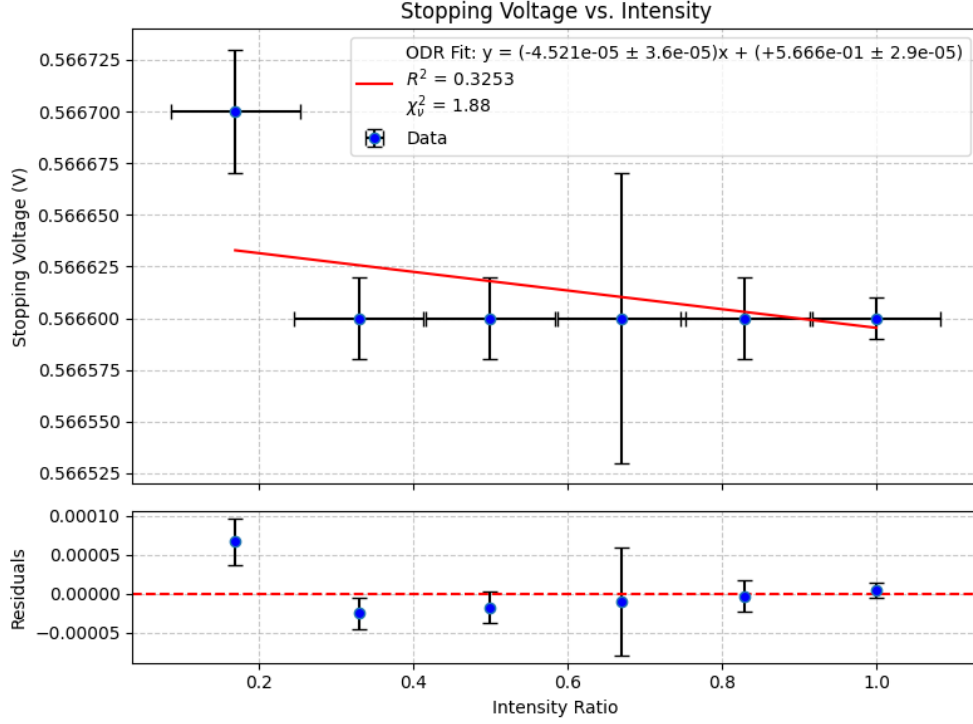


Figure 5: A graph of stopping voltage [V] versus Intensity Ratio i , based on Table 4, with a corresponding plot of residuals. Error bars are present but may not be visible. The linear regression is $V_{stop} = [(-4.5 \pm 0.4) \times 10^{-5}V]i + [(5.66641 \pm 0.00003)V]$. $R^2 = 0.3253$, implying that the stopping voltage is weakly correlated to light intensity. In addition, $\chi^2_\nu = 1.88 > 1$, which implies that the uncertainties of each data point were slightly underestimated. See the Sources of Error section for possible causes of such discrepancies.

Conclusion

Overall, this experiment demonstrated the photoelectric effect using different LEDs. In Part I, an experimental value of Planck's constant, $h_{exp} = (4.9 \pm 0.4) \times 10^{-34}[Js]$ is determined, with the correct magnitude and 26% lower than the accepted value. In addition, the work function $E_{0_{exp}} = (1.4 \pm 0.2) \times 10^{-19}[J]$ and the threshold frequency $f_0 = (2.9 \pm 0.5) \times 10^{14}[Hz]$ are determined. In Part II, using a Green LED and varying intensities, a near constant stopping voltage V_{stop} in changing Intensity i is demonstrated. A strong linear correlation between Photocurrent and Intensity is demonstrated, $I_{photo} = [(6 \pm 2) \times 10^{-21}As]f + [(-2.2 \pm 0.6) \times 10^{-6}A]$. Reinforcing the fact that V_{stop} is independent of Intensity while I_{photo} is linearly correlated to intensity.

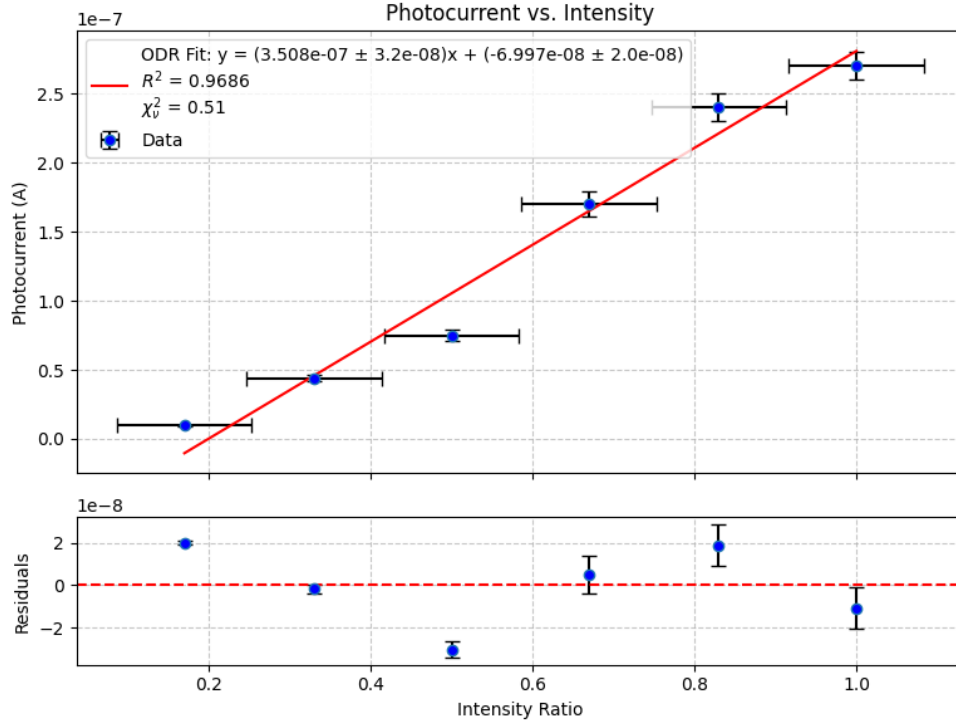


Figure 6: A graph of photocurrent [A] versus Intensity Ratio i , based on Table 5, with a corresponding plot of residuals. Error bars are present but may not be visible. The linear regression is $I_{photo} = [(3.5 \pm 0.3) \times 10^{-7} A]i + [(-7 \pm 2) \times 10^{-8} A]$. $R^2 = 0.9686$, implying that photocurrent is strongly correlated to light intensity. In addition, $\chi^2_\nu = 0.51 < 1$, which implies that the uncertainties of each data point were overestimated. See the Sources of Error section for possible causes of such discrepancies.

References

- [1] Philip R. Bevington and D. Keith Robinson. *Data Reduction and Error Analysis for the Physical Sciences*. McGraw-Hill, 3rd edition, 2003.
- [2] C. Lee, T. Vahabi, and R. M. Serbanescu, “Photoelectric Effect,” Dept. Phys., Univ. Toronto, Toronto, ON, Canada, Lab Manual, Oct. 2025. [Online]. Available: https://www.physics.utoronto.ca/~phy224_324/LabManuals/PhotoelectricEffect.pdf. [Accessed: Mar. 16, 2026].
- [3] OpenStax, “6.3: Photoelectric Effect,” *Physics LibreTexts*, Oct. 23, 2025. [Online]. Available: [https://phys.libretexts.org/Bookshelves/University_Physics/University_Physics_\(OpenStax\)/University_Physics_III_-_Optics_and_Modern_Physics_\(OpenStax\)/06:_Photons_and_Matter_Waves/6.03:_Photoelectric_Effect](https://phys.libretexts.org/Bookshelves/University_Physics/University_Physics_(OpenStax)/University_Physics_III_-_Optics_and_Modern_Physics_(OpenStax)/06:_Photons_and_Matter_Waves/6.03:_Photoelectric_Effect). [Accessed: Mar. 16, 2026].